SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:		Reg. No.:	192425351
Date:		Duration:	60 minutes

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.
2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)

Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4

Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

1. Introduction

This document presents the complete test case design for the Satellite Data Visualization Platform, a web-based system intended to transform raw satellite imagery from public sources such as NASA and NOAA into interactive maps, time-series charts, and anomaly-detection views for meteorologists, disaster-management agencies, and non-expert stakeholders. The purpose of this document is to define the functional and non-functional requirements to be verified, identify test scenarios across every major module, apply recognised test design techniques, and provide a complete, traceable set of test cases covering normal, boundary, invalid, and exceptional conditions.

The scope of testing covers eight core modules derived from the assigned problem statement: satellite data ingestion, interactive map visualization, time-series analysis, anomaly/pattern detection, comparative layer overlay, data export, region-specific filtering, and the general web interface (usability, accessibility, compatibility, and baseline concurrency). Performance and security aspects are covered wherever they can be validated at the functional level; full-scale load and penetration testing are noted as recommended follow-on activities outside the scope of manual functional test cases.

2. Testing Methodology

The test cases in this document were derived using a structured, requirement-driven process, illustrated below. Requirements are first extracted from the problem statement, translated into testable scenarios, and then matched against the most appropriate design technique — Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table (DT), or State Transition Testing (STT) — before individual test cases are written, mapped back to requirements for traceability, and finally executed and re-tested as defects are fixed.

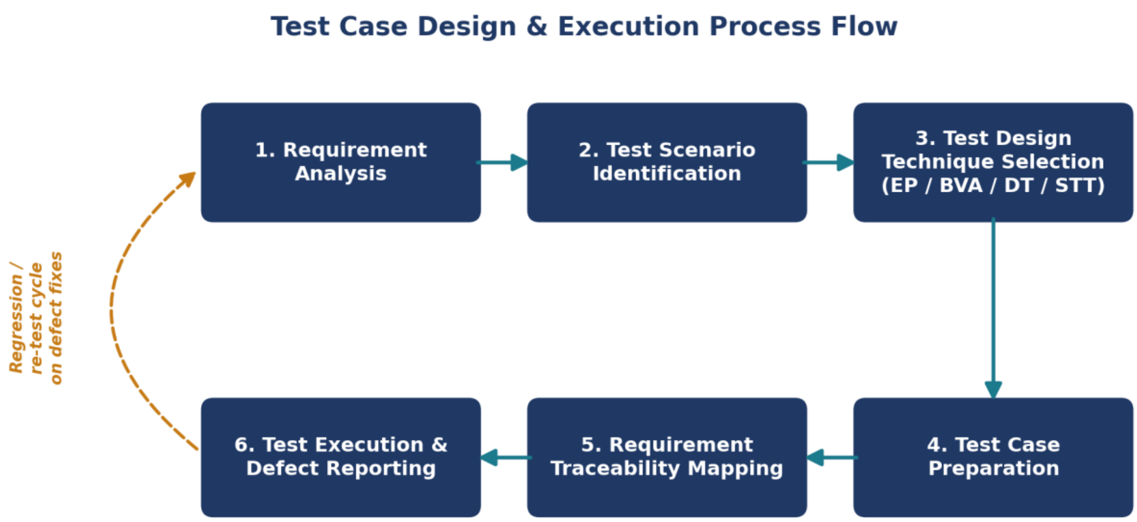

Figure 1: Test Case Design & Execution Process Flow

3. Requirement Identification

3.1 Functional Requirements

Req. ID	Functional Requirement
FR1	The system shall ingest and process satellite imagery data from publicly available sources such as NASA and NOAA.
FR2	The system shall provide an interactive map-based visualization module displaying real-time and historical weather patterns.
FR3	The system shall provide time-series analysis tools to track climate trends such as temperature, rainfall, and cloud cover over a selected period.
FR4	The system shall provide a pattern-detection feature that highlights anomalies such as cyclone formation, droughts, or unusual temperature spikes.
FR5	The system shall provide a comparative visualization tool allowing users to overlay multiple data layers, such as precipitation and vegetation indices.
FR6	The system shall provide data export functionality so researchers can download processed datasets and visualizations.
FR7	The system shall provide a region-specific filtering system allowing users to focus on localized weather and climate data.
FR8	The system shall be accessible through a simple web interface requiring minimal technical expertise to operate.

3.2 Non-Functional Requirements

Req. ID	Category	Non-Functional Requirement
NFR1	**Performance**	Map and chart visualizations shall render within 3 seconds for a standard query under normal load.
NFR2	**Usability**	The interface shall be usable by non-expert users (students, planners, local officials) without prior training.
NFR3	**Reliability**	The data ingestion pipeline shall handle source-API failures gracefully without data loss or corruption.
NFR4	**Scalability**	The platform shall remain responsive under concurrent access by multiple simultaneous users.
NFR5	**Security**	The system shall restrict data export and administrative functions to authenticated users only.
NFR6	**Compatibility**	The web interface shall function correctly across major browsers (Chrome, Firefox, Edge, Safari).
NFR7	**Data Accuracy**	Processed visualizations shall accurately reflect source satellite data within an acceptable margin of error.
NFR8	**Accessibility**	The interface shall be operable via keyboard navigation, consistent with WCAG 2.1 AA guidelines.

4. Test Scenarios

The functional and non-functional requirements above translate into the following ten test scenarios, each covering a distinct module or cross-cutting concern of the platform.

Scenario ID	Module / Feature	Scenario Description	Related Req.
TS-01	**Data Ingestion Pipeline**	Verify satellite imagery is correctly fetched, validated, and processed from public sources.	FR1, NFR3, NFR7
TS-02	**Interactive Map Visualization**	Verify map-based rendering of real-time and historical weather layers.	FR2, NFR1
TS-03	**Time-Series Analysis**	Verify generation of trend charts for temperature, rainfall, and cloud cover over selected periods.	FR3
TS-04	**Anomaly / Pattern Detection**	Verify correct detection and highlighting of climate anomalies.	FR4, NFR7
TS-05	**Comparative Layer Overlay**	Verify overlay and blending of multiple data layers within permitted limits.	FR5
TS-06	**Data Export**	Verify authenticated export of processed datasets in usable formats.	FR6, NFR5
TS-07	**Region-Specific Filtering**	Verify filtering of visualized data to a selected administrative region.	FR7
TS-08	**Usability & Accessibility**	Verify core functions are operable by non-expert users and via keyboard-only navigation.	FR8, NFR2, NFR8
TS-09	**Cross-Browser Compatibility**	Verify consistent rendering across major browsers.	NFR6
TS-10	**Concurrency / Load Smoke Test**	Verify baseline system responsiveness under simultaneous multi-user access.	NFR4

5. Test Design Techniques Applied

5.1 Equivalence Partitioning (EP)

Partition	Description	Sample Input	Expected System Behaviour
Valid class	Date lies within the available satellite-archive range (source start date – current date).	01-06-2020	Date accepted; query proceeds normally.
Invalid class 1	Date is earlier than the archive's earliest available date.	01-01-1950	System rejects the date: “No data available before archive start.”
Invalid class 2	Date is later than the current system date (future date).	01-01-2030	System rejects the date: “Date cannot be in the future.”
Invalid class 3	Input is not a valid date format.	“13/45/2020”	System rejects input and prompts for correct 16/07/2023 format.

Applied to: the date-range input on the Time-Series Analysis module (FR3).

5.2 Boundary Value Analysis (BVA)

sno	Latitude Input	Boundary Type	Expected Result
1	−90	Lower boundary (valid)	Accepted — region at the south-pole boundary renders correctly.
2	−90.0001	Just below lower boundary (invalid)	Rejected with a validation error.
3	0	Nominal / mid-range value	Accepted — equatorial region renders correctly.
4	90	Upper boundary (valid)	Accepted — region at the north-pole boundary renders correctly.
5	90.0001	Just above upper boundary (invalid)	Rejected with a validation error.

Applied to: the latitude field (valid range −90° to +90°) used by region-specific filtering and manual coordinate entry (FR7, TS-02). The same technique applies symmetrically to longitude (−180° to +180°).

5.3 Decision Table Testing

Rule	Precip.	Vegetation	Temp.	Total Layers	Resulting Action
R1	N	N	N	0	No overlay rendered; prompt “Select at least one layer.”
R2	Y	N	N	1	Render single-layer visualization.
R3	Y	Y	N	2	Render combined two-layer overlay.
R4	Y	Y	Y	3 (max)	Render combined three-layer overlay.
R5	Y	Y	Y	4th attempted	Block selection; show “Maximum layer limit (3) reached.”

Applied to: the comparative layer-overlay module, which permits a maximum of three simultaneously selected layers (FR5, TS-05).

5.4 State Transition Testing (STT)

Test ID	Current State	Event / Trigger	Expected Next State
ST-1	Idle	Start ingestion (valid source)	**Fetching**
ST-2	Fetching	Fetch success	**Validating**
ST-3	Fetching	Network timeout / API error	**Failed**
ST-4	Validating	Corrupt data detected	**Failed**
ST-5	Processing	Processing complete	**Ready**
ST-6	Failed	Retry triggered	**Retrying**

Applied to: the data-ingestion pipeline's internal lifecycle (FR1, NFR3), diagrammed below.

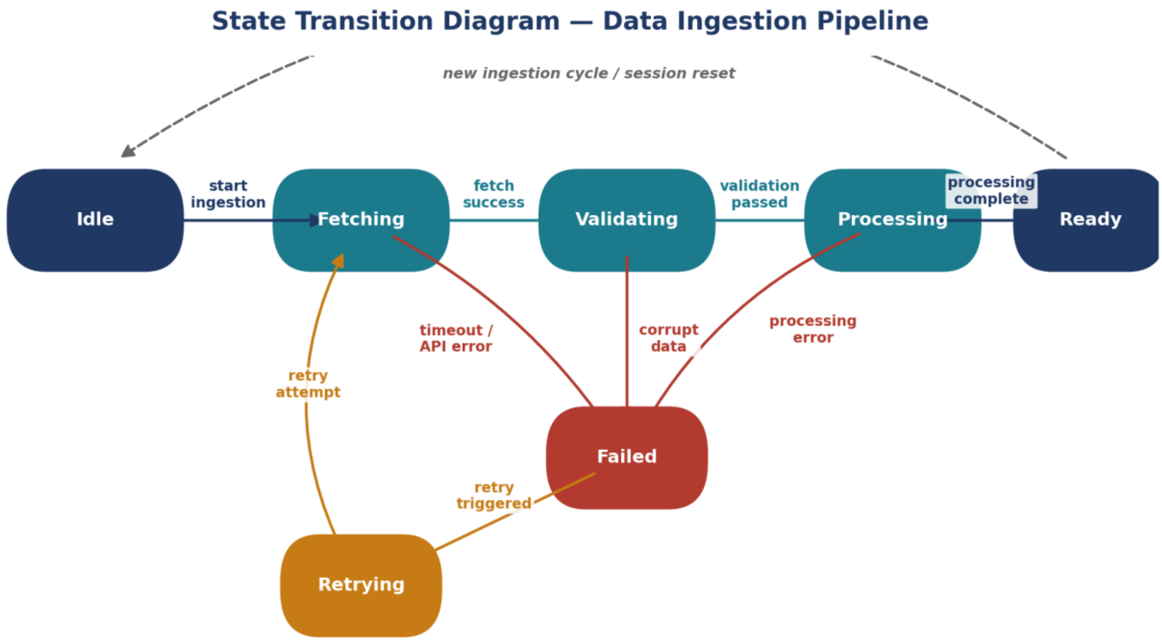

Figure 2: State Transition Diagram — Data Ingestion Pipeline.

6. Detailed Test Cases

The table below lists all 27 test cases, grouped by module, in Normal → Boundary → Invalid → Exceptional order within each group. Each test case is traceable to its parent scenario and requirement via the traceability matrix in Section 7.

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TS-01 — Data Ingestion Pipeline (Related Requirements: FR1, NFR3, NFR7)						
TC -01	**Normal** Ingest valid satellite imagery for a specified date	1. Navigate to the Data Ingestion module. 2. Select source = NASA MODIS, date = 15-Jun-2024. 3. Trigger the “Fetch & Ingest” action. 4. Observe the pipeline status and stored dataset.	Source: NASA MODIS; Date: 2024-06-15; Format: HDF	Imagery is fetched, validated, and processed; dataset status becomes “Ready”; data matches source metadata within tolerance.		
TC -02	**Boundary** Ingest data for the earliest available boundary date	1. Navigate to the Data Ingestion module. 2. Select source = NOAA AVHRR, date = archive's earliest date. 3. Trigger ingestion.	Date = 01-11-1978 (lower boundary of archive)	System accepts the boundary date and completes ingestion successfully; the valid boundary is not incorrectly excluded.		
TC -03	**Invalid** Attempt ingestion of an unsupported file format	1. Navigate to the Data Ingestion module. 2. Supply a file with extension “.txt” instead of “.hdf”/“.nc”. 3. Trigger ingestion.	File = sample_data.txt	System rejects the file with a clear “Unsupported format” error; pipeline does not crash; no partial data is stored.		
TC -04	**Exceptional** Source API unreachable during fetch	1. Simulate NOAA API downtime / network timeout. 2. Trigger ingestion for a valid date. 3. Observe system behaviour and logs.	Source = NOAA (simulated timeout / 503)	Failure is caught and logged; dataset status becomes “Failed”; a retry option is offered; system does not crash (NFR3).		
TS-02 — Interactive Map Visualization (Related Requirements: FR2, NFR1)						
TC -05	**Normal** Load map with a current cloud-cover overlay	1. Open the Map Visualization module. 2. Select region = Tamil Nadu; layer = Cloud Cover. 3. Load the map.	Region = Tamil Nadu; Layer = Cloud Cover; Date = today	Map renders with the correct overlay within 3 seconds (NFR1); legend is accurate.		
TC -06	**Boundary** Zoom to the maximum supported zoom level	1. Load the map for any valid region. 2. Zoom in repeatedly to reach the maximum zoom level (18).	Zoom level = 18 (max)	Tiles render correctly at maximum zoom; no layout breakage or missing tiles.		
TC -07	**Invalid** Enter an out-of-range coordinate	1. Open manual coordinate entry. 2. Enter latitude = 200, longitude = 80. 3. Submit.	Latitude = 200 (invalid; valid range −90 to 90)	System shows “Latitude must be between −90 and 90” and does not attempt to render.		
TC -08	**Exceptional** Map tile server times out mid-render	1. Simulate a slow / timing-out tile server. 2. Load the map for a valid region.	Tile server response = timeout (simulated)	A fallback error message / placeholder is shown instead of an indefinite blank screen; no crash.		
TS-03 — Time-Series Analysis (Related Requirements: FR3)						

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC -09	**Normal** Generate a temperature time-series chart for a valid range	1. Open the Time-Series module. 2. Select parameter = Temperature; region = Chennai; range = 01-Jan-2023 to 31-Dec-2023. 3. Generate the chart.	Range = 2023-01-01 to 2023-12-31	Chart renders accurately with 12 months of data reflecting the seasonal trend.		
TC -10	**Boundary** Select the minimum date range (single day)	1. Open the Time-Series module. 2. Set start date = end date = 15-Mar-2024. 3. Generate the chart.	Range = 2024-03-15 to 2024-03-15	Chart renders a single valid data point without error.		
TC -11	**Boundary** Select the maximum permitted date range	1. Open the Time-Series module. 2. Select the full historical archive span. 3. Generate the chart.	Range = full archive span (earliest date – today)	System aggregates data appropriately (e.g., yearly averages) and renders within an acceptable time, or shows a clear limit message — no timeout crash (NFR1).		
TC -12	**Invalid** Start date entered later than end date	1. Open the Time-Series module. 2. Enter start date = 01-Jan-2024, end date = 01-Jan-2023. 3. Submit.	Start date > end date	System shows “Start date must be before end date”; chart is not generated.		
TC -13	**Invalid** Non-date formatted string entered	1. Open the Time-Series module. 2. Enter start date = “abcd”. 3. Submit.	Start date = “abcd”	Malformed input is rejected; user is prompted for the correct DD-MM-YYYY format.		
TS-04 — Anomaly / Pattern Detection (Related Requirements: FR4, NFR7)						
TC -14	**Normal** Detect a known cyclone signature	1. Load a labelled historical dataset containing a documented cyclone event. 2. Run anomaly detection.	Dataset = reference cyclone event (documented)	System correctly flags and highlights the cyclone formation matching the known event window (NFR7).		
TC -15	**Normal** Run detection on a stable dataset (no anomalies)	1. Load a dataset with normal, stable weather conditions. 2. Run detection.	Dataset = stable region / period, no known events	System returns “No anomalies detected”; no false positives are raised.		
TC -16	**Exceptional** Run detection on an incomplete / corrupted dataset	1. Load a dataset with deliberately missing / corrupted frames. 2. Run detection.	Dataset = corrupted (30% of frames missing)	System reports a data-quality warning instead of producing an unreliable anomaly result.		
TS-05 — Comparative Layer Overlay (Related Requirements: FR5)						
TC -17	**Normal** Overlay two compatible layers	1. Open the Comparative Visualization module. 2. Select layers = Precipitation + Vegetation Index. 3. Render.	Layers = {Precipitation, Vegetation Index}	Both layers overlay correctly with accurate blending and legend (Decision Table rule R3).		
TC -18	**Boundary** Select the maximum	1. Select 3 layers: Precipitation, Vegetation Index, Temperature. 2. Render.	Layers = 3 (system maximum)	All three layers render together correctly without performance degradation		

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
	allowed number of layers			(rule R4).		
TC-19	**Invalid** Attempt to add a 4th layer beyond the maximum	1. With 3 layers already selected, attempt to add a 4th.	Layers = 4 (exceeds maximum)	System blocks the 4th selection and shows “Maximum layer limit (3) reached” (rule R5).		
TS-06 — Data Export (Related Requirements: FR6, NFR5)						
TC-20	**Normal** Export a processed dataset as CSV while authenticated	1. Log in as a registered user. 2. Open the Export module; select dataset and format = CSV. 3. Trigger export.	Format = CSV; user = authenticated	File downloads successfully; exported data matches the visualized dataset.		
TC-21	**Invalid** Attempt export without authentication	1. Access the Export module without logging in. 2. Attempt to trigger export.	User = not authenticated	System denies the action and redirects to the login page (NFR5).		
TC-22	**Exceptional** Export a dataset near the maximum permitted file size	1. Select a dataset close to the platform's maximum export size. 2. Trigger export.	File size ≈ configured maximum (e.g., 500 MB)	Export completes with a progress indicator, or is gracefully denied with a clear size-limit message — no server timeout or crash.		
TS-07 — Region-Specific Filtering (Related Requirements: FR7)						
TC-23	**Normal** Filter visualization by a valid administrative region	1. Open the filter panel. 2. Select region = Karnataka. 3. Apply the filter.	Region = Karnataka	Only data relevant to the selected region is displayed; other regions are excluded.		
TC-24	**Invalid** Enter a non-existent region name	1. Open the filter panel; manually type region = “Atlantis”. 2. Apply the filter.	Region = “Atlantis” (non-existent)	System shows “Region not found” and displays an empty state gracefully.		
TS-08 — Usability & Accessibility (Related Requirements: FR8, NFR2, NFR8)						
TC-25	**Normal** Operate all core functions using keyboard navigation only	1. Avoid mouse use for the duration of the test. 2. Use Tab / Enter / Arrow keys to select a region, choose a layer, and generate a chart.	Input method = keyboard only	All core functions are reachable and operable via keyboard, consistent with WCAG 2.1 AA (NFR8).		
TS-09 — Cross-Browser Compatibility (Related Requirements: NFR6)						
TC-26	**Normal** Verify rendering across major browsers	1. Open the platform in Chrome, Firefox, Edge, and Safari. 2. Load the same map visualization in each.	Browsers = Chrome, Firefox, Edge, Safari (latest stable)	Visualization renders consistently and correctly across all four browsers (NFR6).		

ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TS-10 — Concurrency / Load Smoke Test (Related Requirements: NFR4)						
TC-27	**Exceptional** Simulate concurrent access by multiple users	1. Use a load-simulation tool to simulate 50 simultaneous map-visualization requests. 2. Observe system responsiveness and error rate.	Concurrent users = 50 (functional smoke test; full 500-user load test recommended separately)	System remains responsive with no failed requests or crashes at this scale, confirming baseline concurrency handling (NFR4).		

Note: "Actual Result" and "Status" (Pass / Fail) are intentionally left blank in this design document — they are to be completed by the tester during the test-execution phase, once the platform build under test is available.

7. Requirement Traceability Matrix

All 8 functional requirements and all 8 non-functional requirements defined in Section 3 are mapped to at least one test case, giving 100% requirement coverage across the 27 test cases designed in Section 6.

Req. ID	Type	Requirement Summary	Related Test Case(s)	Coverage Status
FR1	**FR**	Ingest & process satellite imagery from NASA/NOAA	TC-01, TC-02, TC-03, TC-04	**Covered**
FR2	**FR**	Interactive map-based visualization of weather patterns	TC-05, TC-06, TC-07, TC-08	**Covered**
FR3	**FR**	Time-series analysis of temperature, rainfall, cloud cover	TC-09, TC-10, TC-11, TC-12, TC-13	**Covered**
FR4	**FR**	Pattern/anomaly detection (cyclones, droughts, spikes)	TC-14, TC-15, TC-16	**Covered**
FR5	**FR**	Comparative multi-layer overlay visualization	TC-17, TC-18, TC-19	**Covered**
FR6	**FR**	Data export functionality for researchers	TC-20, TC-21, TC-22	**Covered**
FR7	**FR**	Region-specific filtering	TC-23, TC-24	**Covered**
FR8	**FR**	Simple, low-expertise web interface	TC-25	**Covered**
NFR1	**NFR**	Performance — render within 3 seconds	TC-05, TC-11	**Covered**
NFR2	**NFR**	Usability — operable without training	TC-25	**Covered**
NFR3	**NFR**	Reliability — graceful handling of failures	TC-04, TC-16	**Covered**
NFR4	**NFR**	Scalability — concurrent multi-user access	TC-27	**Covered (baseline)[1]**
NFR5	**NFR**	Security — authenticated export/admin access	TC-21	**Covered**
NFR6	**NFR**	Compatibility — major browsers	TC-26	**Covered (baseline)[1]**
NFR7	**NFR**	Data accuracy vs. source imagery	TC-01, TC-14	**Covered**
NFR8	**NFR**	Accessibility — WCAG 2.1 AA / keyboard nav	TC-25	**Covered**

[1] Baseline functional smoke test only; full-scale load testing (≈500 concurrent users, via tools such as JMeter) and a dedicated cross-browser/device compatibility pass are recommended as follow-on non-functional testing activities.

8. Test Coverage Summary

The 27 designed test cases are distributed across four condition categories — Normal, Boundary, Invalid, and Exceptional — to ensure the platform is verified not only under expected use but also at its operating limits and under failure conditions. The chart below shows this distribution, followed by the depth of coverage achieved for each functional requirement.

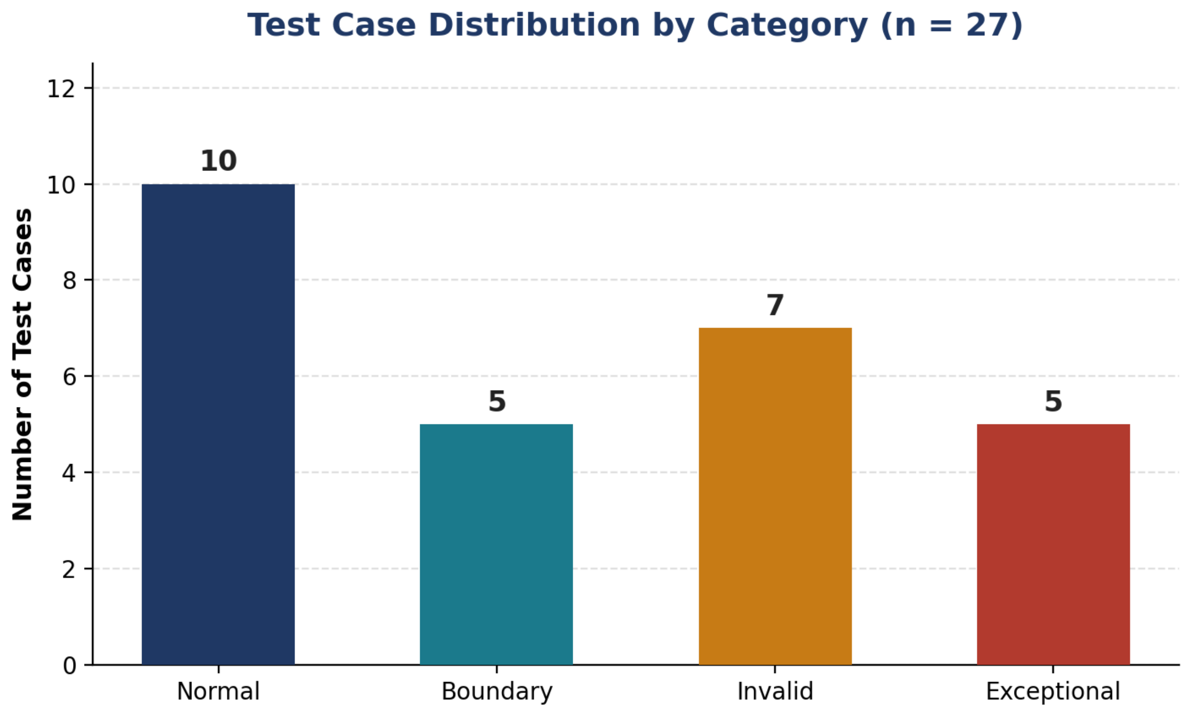

Figure 3: Test Case Distribution by Category

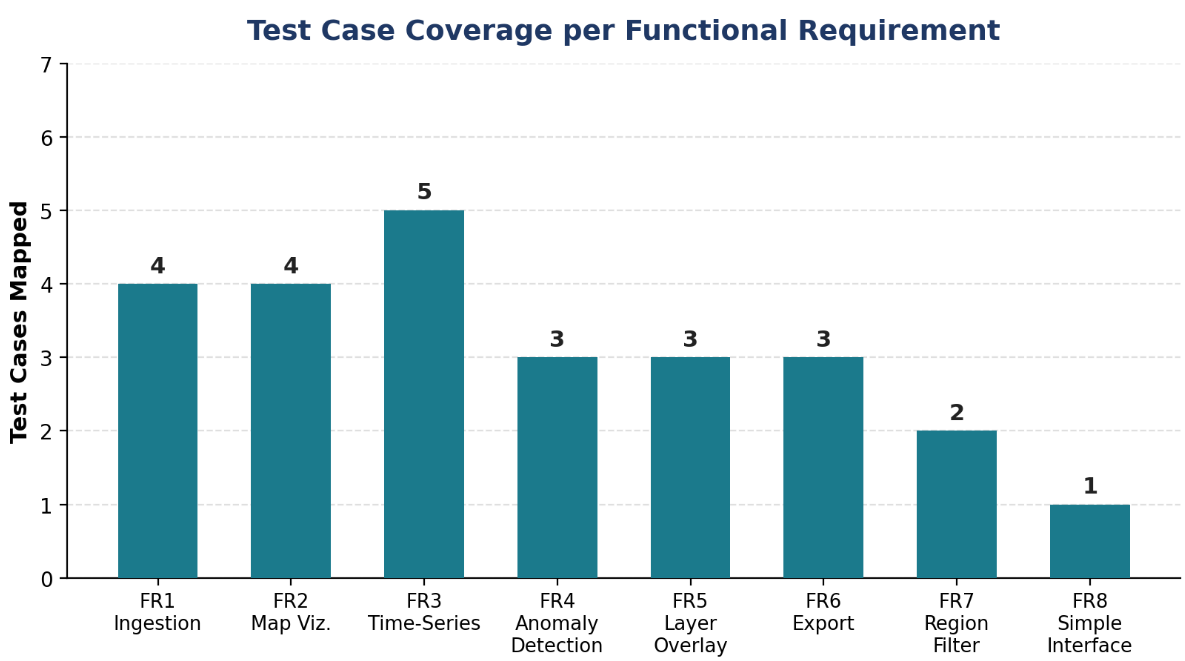

Figure 4: Test Case Coverage per Functional Requirement

9. Conclusion

This document has identified 8 functional and 8 non-functional requirements from the Satellite Data Visualization Platform problem statement, derived 10 test scenarios spanning every core module, applied four recognised test design techniques (Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition Testing), and produced 27 detailed, traceable test cases covering normal, boundary, invalid, and exceptional conditions.

The accompanying Requirement Traceability Matrix confirms 100% mapping between requirements and test cases, giving reasonable confidence that, once executed, this test suite will surface defects in data ingestion reliability, visualization correctness, input validation, security, and accessibility before the platform reaches meteorologists, disaster-management agencies, and other end users. Non-functional areas that require specialised tooling — full-scale load testing and a dedicated compatibility matrix — are flagged for a follow-on testing phase beyond this functional test case design.